

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403252
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Kamen; Dean et al.

Infusion device system and apparatus

Abstract

An infusion device. The infusion device includes a top portion comprising an introduction needle, and a bottom portion comprising a cannula, the top portion removably attached to the bottom portion, wherein the introduction needle is used to insert the cannula, and wherein after insertion, the top portion is removed from the bottom portion.

Inventors:	Kamen; Dean (Bedford, NH), Grant; Kevin L. (Litchfield, NH)
Applicant:	DEKA Products Limited Partnership (Manchester, NH)
Family ID:	55272697
Assignee:	DEKA PRODUCTS LIMITED PARTNERSHIP (Manchester, NH)
Appl. No.:	17/878217
Filed:	August 01, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220362459 A1	Nov. 17, 2022

Related U.S. Application Data

continuation parent-doc US 16572032 20190916 US 11400209 child-doc US 17878217
continuation parent-doc US 14996559 20160115 US 10413661 child-doc US 16572032
us-provisional-application US 62103841 20150115

Publication Classification

Int. Cl.: A61M5/162 (20060101); A61M5/14 (20060101); A61M5/142 (20060101); A61M5/158 (20060101)

U.S. Cl.:

CPC **A61M5/162** (20130101); **A61M5/1413** (20130101); **A61M5/14248** (20130101);
A61M5/158 (20130101); A61M2005/14268 (20130101); A61M2005/1585 (20130101);
A61M2005/1588 (20130101); A61M2205/0266 (20130101); A61M2205/3337
(20130101); A61M2205/3379 (20130101); A61M2205/3389 (20130101); A61M2205/36
(20130101); A61M2205/582 (20130101); A61M2205/8206 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4755173	12/1987	Konopka	604/122	A61M 25/0606
6685674	12/2003	Douglas et al.	N/A	N/A
8083730	12/2010	Miesel	N/A	N/A
8303549	12/2011	Mejlhede	604/200	A61M 5/14248
10413661	12/2018	Kamen et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO/2009/016638	12/2008	WO	N/A
WO/2009/125398	12/2008	WO	N/A
WO/2010/113159	12/2009	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability for International Application No.

PCT/US2016/013603 dated Jul. 18, 2017. cited by applicant

International Search Report for International Application No. PCT/US2016/013603 dated Jul. 4, 2017. cited by applicant

Written Opinion of the International Searching Authority for International Application No. PCT/US2016/013603 dated Jul. 4, 2017. cited by applicant

Primary Examiner: Mendez; Manuel A

Attorney, Agent or Firm: McCormick, Paulding & Huber PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) (1) The present application is a continuation of U.S. patent application Ser. No. 16/572,032, filed Sep. 16, 2019, which is a continuation of U.S. patent application Ser. No. 14/996,559, filed Jan. 15, 2016, now U.S. Pat. No.

10,413,661, issued Sep. 17, 2019 and entitled Apparatus, System and Method for Fluid Delivery which is a Non-provisional Application which claims priority from U.S. Provisional Patent Application Ser. No. 62/103,841, filed Jan. 15, 2015 and entitled Infusion Pump Assembly, which is hereby incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) This application relates generally to fluid delivery systems, and more particularly to an infusion pump assemblies and infusion devices, systems and apparatus.

BACKGROUND

(2) Many potentially valuable medicines or compounds, including biologicals, are not orally active due to poor absorption, hepatic metabolism or other pharmacokinetic factors. Additionally, some therapeutic compounds, although they can be orally absorbed, are sometimes required to be administered so often it is difficult for a patient to maintain the desired schedule. In these cases, parenteral delivery is often employed or could be employed.

(3) Effective parenteral routes of drug delivery, as well as other fluids and compounds, such as subcutaneous injection, intramuscular injection, and intravenous (IV) administration include puncture of the skin with a needle or stylet. Insulin is an example of a therapeutic fluid that is self-injected by millions of diabetic patients. Users of parenterally delivered drugs may benefit from a wearable device that would automatically deliver needed drugs/compounds over a period of time.

(4) To this end, there have been efforts to design portable and wearable devices for the controlled release of therapeutics. Such devices are known to have a reservoir such as a cartridge, syringe, or bag, and to be electronically controlled. These devices suffer from a number of drawbacks including the malfunction rate. Reducing the size, weight and cost of these devices is also an ongoing challenge. Additionally, these devices often apply to the skin and pose the challenge of frequent re-location for application.

SUMMARY OF THE INVENTION

(5) In accordance with first implementation, an infusion device system is disclosed. The infusion device system includes a disposable housing assembly comprising an infusion device mating assembly attached to the disposable housing assembly comprising a piercing needle and a reservoir, wherein the piercing needle fluidly connected to the reservoir; and an infusion device including a top portion comprising an introduction needle and a bottom portion including a septum and a cannula, the top portion removably attached to the bottom portion. Wherein the introduction needle is used to insert the cannula, and wherein after insertion, the top portion is removed from the bottom portion, wherein the infusion device mating assembly configured to attach to the bottom portion of the infusion device, wherein the piercing needle configured to pierce the septum, and wherein when the infusion device mating assembly is attached to the bottom portion of the infusion device, the cannula is fluidly connected to the reservoir.

(6) Some embodiments of this implementation include one or more of the following. Wherein the system further includes a predetermined length of tubing comprising a first end and a second end. Wherein the first end of the tubing configured to attach to the bottom portion of the infusion device and the second end of the tubing configured to attach to the infusion device mating assembly. Wherein the second end of the tubing configured to attach to the piercing needle, wherein the tubing is fluidly connected to the reservoir. Wherein the system further includes an autoinsertor. Wherein the system further includes a reusable housing assembly configured to removably attach to the disposable housing assembly. Wherein the reusable housing assembly includes a volume sensor assembly. Wherein the disposable housing assembly includes a pumping chamber.

(7) In accordance with another implementation, an infusion device system is disclosed. The infusion device system includes a disposable housing assembly including an infusion device mating assembly attached to the disposable housing assembly comprising a piercing needle and a reservoir, wherein the piercing needle fluidly connected to the reservoir. The system also includes a

reusable housing assembly configured to removably attach to the disposable housing assembly and an infusion device. The infusion device includes a top portion including an introduction needle and a bottom portion including a septum and a cannula, the top portion removably attached to the bottom portion. Wherein the introduction needle is used to insert the cannula, and wherein after insertion, the top portion is removed from the bottom portion, wherein the infusion device mating assembly configured to attach to the bottom portion of the infusion device, wherein the piercing needle configured to pierce the septum, and wherein when the infusion device mating assembly is attached to the bottom portion of the infusion device, the cannula is fluidly connected to the reservoir.

(8) Some embodiments of this implementation include one or more of the following. Wherein the infusion device system further includes a predetermined length of tubing including a first end and a second end. Wherein the first end of the tubing configured to attach to the bottom portion of the infusion device and the second end of the tubing configured to attach to the infusion device mating assembly. Wherein the second end of the tubing configured to attach to the piercing needle, wherein the tubing is fluidly connected to the reservoir. Wherein the system further including an autoinsertor. Wherein the reusable housing assembly including a volume sensor assembly. Wherein the disposable housing assembly including a pumping chamber.

(9) In accordance with first implementation, an infusion device is disclosed. The infusion device includes a top portion including an introduction needle and a bottom portion comprising a septum and a cannula, the top portion removably attached to the bottom portion, wherein the introduction needle is used to insert the cannula, and wherein after insertion, the top portion is removed from the bottom portion, wherein the bottom portion configured to attach to an infusion device mating assembly of an infusion pump assembly, wherein when the infusion device mating assembly is attached to the bottom portion of the infusion device, the cannula is fluidly connected to a reservoir.

(10) Some embodiments of this implementation include one or more of the following. The infusion device further including a predetermined length of tubing comprising a first end and a second end. Wherein the first end of the tubing configured to attach to the bottom portion of the infusion device and the second end of the tubing configured to attach to the infusion device mating assembly. The infusion device further including an autoinsertor. Wherein the bottom portion including a septum.

(11) The details of one or more embodiments are set forth in the accompanying drawings and the description below. Other features and advantages will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A-3C are various views of one embodiment of an infusion device system;
- (2) FIGS. 4A-4K are various views of one embodiment of an infusion device system;
- (3) FIGS. 5A-5F are various views of one embodiment of an infusion device system;
- (4) FIGS. 6A-6G are various views of one embodiment of an infusion device system; and
- (5) FIG. 7A is an exploded view of various components of an infusion pump assembly;
- (6) FIG. 7B is an isometric view of a portion of the infusion pump assembly;
- (7) FIG. 8 is a cross-sectional view of an embodiment of a disposable housing assembly of an embodiment of an infusion pump assembly;
- (8) FIG. 9 is a diagrammatic view of a fluid path within an embodiment of the infusion pump assembly; and
- (9) FIG. 10 is a view of another embodiment of an infusion device system including a tubing.
- (10) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

(11) In various embodiments, an infusion device and system and methods thereof is disclosed. The infusion device is configured to be inserted into the subcutaneous layer of a user's skin and be fluidly connected to a fluid source. In various embodiments, the infusion device may be fluidly connected to a length of tubing and/or to an infusion pump. Infusion pumps include any infusion pump which may include, but is not limited to, the various infusion pumps shown and described in U.S. patent application Ser. No. 13/788,260, filed Mar. 7, 2013 and entitled Infusion Pump Assembly, now U.S. Publication No. US-2014-0107579, published Apr. 17, 2014; U.S. Pat. No. 8,491,570, issued Jul. 23, 2013 and entitled Infusion Pump Assembly; U.S. Pat. No. 8,414,522, issued Apr. 9, 2013 and entitled Fluid Delivery Systems and Methods; U.S. Pat. No. 8,262,616, issued Sep. 11, 2012 and entitled Infusion Pump Assembly; and U.S. Pat. No. 7,306,578, issued Dec. 11, 2007 and entitled Loading Mechanism for Infusion Pump; all of which are hereby incorporated herein by reference in their entireties. In various embodiments, the various embodiments of the infusion devices described herein may be used alone or in conjunction with an autoinsertor.

(12) In various embodiments, an infusion device is disclosed that includes an introduction needle and a cannula but, in various embodiments, includes embodiments that do not include additional tubing connected to a fluid source. Rather, in these embodiments, the cannula is directly fluidly connected to a fluid source. This may be desirable/beneficial for many reasons, including but not limited to, the ability to infuse fluids subcutaneously without the hassles and inconvenience of additional tubing which has various downsides including, but not limited to, being caught, getting tangled, becoming occluded, and difficulty in concealing the tubing for everyday use. However, in some embodiments, the system described herein includes the option of fluidly attaching tubing to the cannula. Additionally, some embodiments include an autoinsertor while other embodiments do not. Embodiments without an autoinsertor may be beneficial/desirable for many reasons, including, but not limited to, the size of the infusion device may be minimized and the simplicity of the infusion device may minimize costs to both manufacturers and users.

(13) Referring now to FIGS. 1A-3C, in various embodiments, the infusion device system **108** may include an infusion device **106** which in various embodiments may include a top portion **110**, a bottom portion **112**, a cannula **114**, an introduction needle **116** and an adhesive strip **118**. The introduction needle **116** is attached to the top portion **110** and the cannula **114** is attached to the bottom portion **112**. In various embodiments, the top portion **110** may be configured such that it is easily maneuvered by a user and/or caregiver. Although the size and shape of the top portion **110** may vary in various embodiments, in some embodiments, the size and shape are ergonomic and easily maneuverable to complete the insertion of the cannula **114** by a user and/or caregiver. In various embodiments, inside the bottom portion **112** is a septum **122** in which the introduction needle pierces through in the starting position, which is shown in, for example, FIGS. 1A-1C. In various embodiments, the infusion device system **108** may also include a fluid source which in some embodiments may be an infusion pump assembly/apparatus which may include a disposable housing assembly **102** and a reusable housing assembly **100**. In various embodiments, the disposable housing assembly **102** and reusable housing assembly **100** are configured to be removably attached to one another.

(14) Referring now also to FIGS. 7A-7B & 8-9, there are shown various views of infusion pump assembly **500**, which is shown to include reusable housing assembly **502**, switch assembly **506**, and main body portion **508**. In various embodiments, main body portion **508** may include a plurality of components, examples of which may include but are not limited to a volume sensor assembly **548**, printed circuit board **600**, vibration motor assembly **602**, shape memory actuator anchor **604**, switch assembly **506**, battery **606**, antenna assembly **608**, pump assembly **106**, measurement valve assembly **610**, volume sensor valve assembly **612** and reservoir valve assembly **614**. To enhance clarity, printed circuit board **600** has been removed from FIG. 7B to allow for viewing of the various components positioned beneath printed circuit board **600**.

(15) The various electrical components that may be electrically coupled with printed circuit board **600** may utilize spring-biased terminals that allow for electrical coupling without the need for soldering the connections. For example, vibration motor assembly **602** may utilize a pair of spring-biased terminals (one positive terminal and one negative terminal) that are configured to press against corresponding conductive pads on printed circuit board **600** when vibration motor assembly **602** is positioned on printed circuit board **600**. However, in the exemplary embodiment, vibration motor assembly **602** is soldered directly to the printed circuit board.

(16) The volume sensor assembly **548** may be configured to monitor the amount of fluid infused by infusion pump assembly **500**. For example, volume sensor assembly **548** may employ acoustic volume sensing, which is the subject of U.S. Pat. Nos. 5,575,310 and 5,755,683 assigned to DEKA Products Limited Partnership, as well as the U.S. Pat. No. 8,414,522, issued Apr. 9, 2013, and entitled Fluid Delivery Systems and Methods; U.S. Pat. No. 8,545,445, issued Oct. 1, 2013, and entitled Patch-Sized Fluid Delivery Systems and Methods and U.S. Pat. No. 8,113,244, issued Feb. 14, 2012, and entitled Adhesive and Peripheral Systems and Methods for Medical Devices, the entire disclosures of all of which are incorporated herein by reference.

(17) Vibration motor assembly **602** may be configured to provide a vibration-based signal to the user of infusion pump assembly **500**. For example, in the event that the voltage of battery **606** (which powers infusion pump assembly **500**) is below the minimum acceptable voltage, vibration motor assembly **602** may vibrate infusion pump assembly **500** to provide a vibration-based signal to the user of infusion pump assembly **500**. Shape memory actuator anchor **604** may provide a mounting point for the above-described shape memory actuator (e.g. shape memory actuator **512**). In various embodiments, shape memory actuator **512** may be, for example, a conductive shape-memory alloy wire that changes shape with temperature. The temperature of shape-memory actuator **512** may be changed with a heater, or more conveniently, by application of electrical energy. Accordingly, one end of shape memory actuator **512** may be rigidly affixed (i.e., anchored) to shape memory actuator anchor **604** and the other end of shape memory actuator **512** may be applied to e.g. a valve assembly and/or a pump actuator. Therefore, by applying electrical energy to shape memory actuator **512**, the length of shape memory actuator **512** may be controlled and, therefore, the valve assembly and/or the pump actuator to which it is attached may be manipulated. Various methods and systems related to the control and actuation of shape memory actuator **512** and shape memory alloys are described in U.S. patent application Ser. No. 12/837,193, filed Jul. 15, 2010 and entitled Apparatus, Systems and Methods for an Infusion Pump Assembly, now U.S. Publication No. US-2011-0144574, published Jun. 16, 2011; U.S. Pat. No. 8,852,164, issued Oct. 7, 2014, and entitled Method and System for Shape-Memory Alloy Wire Control and U.S. Pat. No. 8,579,884, issued Nov. 12, 2013 and entitled Infusion Pump Assembly each of which is incorporated herein by reference in its entirety.

(18) Antenna assembly **608** may be configured to allow for wireless communication between e.g. infusion pump assembly **500** and a remote control assembly (not shown). The remote control assembly may allow the user to program infusion pump assembly **500** and e.g. configure bolus and basal infusion events. As discussed above, infusion pump assembly **500** may include one or more valve assemblies configured to control the flow of the infusible fluid through a fluid path **524** (within infusion pump assembly **500**) and pump assembly **526A**, located in the reusable housing assembly **502**, may be configured to pump the infusible fluid from the fluid path **524** to the user, and in various embodiments, pumps fluid from the reservoir, into the pumping chamber **526B** (located in the disposable housing assembly **504**). In this particular embodiment of infusion pump assembly **500**, infusion pump assembly **500** is shown to include three valve assemblies, namely measurement valve assembly **610**, volume sensor valve assembly **612**, and reservoir valve assembly **614**.

(19) As discussed above and referring also to FIG. **9**, the infusible fluid may be stored within reservoir **518**. In order to effectuate the delivery of the infusible fluid to the user, the processing

logic (not shown) included within infusion pump assembly **500** may energize shape memory actuator **512**, which may be anchored on one end using shape memory actuator anchor **604**. Shape memory actuator **512** may result in the activation of pump assembly **526A** and reservoir valve assembly **614**. Reservoir valve assembly **614** may include reservoir valve actuator **614A** and reservoir valve **614B**, and the activation of reservoir valve assembly **614** may result in the downward displacement of reservoir valve actuator **614A** and the closing of reservoir valve **614B**, resulting in the effective isolation of reservoir **518**. Further, pump assembly **526A** may include a pump plunger and pump chamber **526B** and the activation of the pump assembly **526A** may result in pump plunger being displaced in a downward fashion into pump chamber **526B** and the displacement of the infusible fluid (in the direction of arrow **616**).

(20) Volume sensor valve assembly **612** may include volume sensor valve actuator **612A** and volume sensor valve **612B**. In various embodiments, volume sensor valve actuator **612A** may be closed via a spring assembly that provides mechanical force to seal volume sensor valve **612B**. However, when pump assembly **526A** is activated, if the displaced infusible fluid is of sufficient pressure to overcome the mechanical sealing force of volume sensor valve assembly **612**, the displacement of the infusible fluid occurs in the direction of arrow **618**. This may result in the filling of volume sensor chamber **620** included within volume sensor assembly **548**. Through the use of speaker assembly **622**, port assembly **624**, reference microphone **626**, spring diaphragm **628**, invariable volume microphone **630**, volume sensor assembly **548** may determine the volume of infusible fluid included within volume sensor chamber **620**.

(21) In various embodiments, the disposable housing assembly may be any of the embodiment described herein or incorporated by reference. In any one or more of the embodiments of the disposable housing assembly, the disposable housing assembly includes a reservoir. An embodiment of the reservoir is shown in FIG. **8**, however, this embodiment of the reservoir is not limited to the embodiment of the disposable housing assembly shown in FIG. **8**. Rather, the reservoir is included, even if not shown, in all embodiments of the disposable housing assembly. Further, the various embodiments of the disposable housing assembly described herein may be used with any of the various embodiments of the infusion device described herein and therefore, each embodiment of the infusion device system is not limited only to the embodiment of the disposable housing assembly shown and described with respect to that particular embodiment of the infusion device.

(22) Referring also to FIGS. **1A-3C**, in various embodiments, and as discussed above, the disposable housing assembly **102** includes a reservoir **518** and an infusion device mating assembly **104**. The infusion device mating assembly **104** includes a piercing needle **120** which passes through the opening **124** on the bottom portion **112** of the infusion device **106**. The piercing needle **120** is fluidly connected to the reservoir **518**.

(23) In practice, the infusion device **106** is pushed into the subcutaneous skin layer of a user by applying a downward force to the top portion **110** of the infusion device **106**. The introduction needle **116** pierces the skin and provides entry for the cannula **114**. Once this is completed, the top portion **110** may be removed from the bottom portion **112** by applying an opposite force onto the top portion **110**. The top portion **110** and introduction needle **116** are then removed from the bottom portion **112** and the user. The adhesive strip **118** maintains the location of the bottom portion **112** on the skin of the user.

(24) In various embodiments, the adhesive strip **118** may be any size and shape, including, but not limited to, the size and shape shown in FIGS. **1A-3C**.

(25) The disposable housing assembly **102** infusion device mating assembly **104** may be mated with the bottom portion **112** of the infusion device **106** by placing the infusion device mating assembly **104** on top of the bottom portion **112** and the application of force in the direction towards the bottom portion **112**. In various embodiments, the application of force causes a piercing needle **120** to pierce a septum which, in some embodiments, is located in the opening **124**. The application

of force eases the piercing needle **120** through the opening **124** and the piercing needle **120** fluidly connects to the cannula **114**, which, in various embodiments, fluidly connects the reservoir **518** to the cannula **114**. In various embodiments, the infusion device mating assembly **104** may be attached to the bottom portion **112** in any orientation, thus, the user may place the disposable housing assembly **102** in any orientation i.e., within 360 degrees, with respect to the bottom portion **112**.

(26) In various embodiments of this embodiment of the infusion device system **108**, if and when a user wishes to move the location of the disposable housing assembly **102**, the infusion device mating assembly **104** may be removed from the bottom portion **112** and the bottom portion **112** and cannula **114** may be removed from the user. Following, the user may connect another infusion device **106** bottom portion **112** to their skin in a different location on their body and reconnect the disposable housing assembly **102** infusion device mating assembly **104**. In some embodiments, rather than removing the bottom portion **112** and the cannula **114**, the infusion device system may include a tubing or other that connects to the bottom portion **112** on a first end of the tubing and to the infusion device mating assembly **104** on the second end of the tubing, thereby creating a fluid connection between the disposable housing assembly **102** and the cannula **114** without the need for removing a first cannula and inserting a second cannula.

(27) Referring now also to FIGS. **4A-4K**, in various embodiments, the infusion device system **218** may include an infusion device **206** which in various embodiments may include a top portion **210**, a bottom portion **212**, a cannula **214** and an introduction needle **216**. The introduction needle **216** is attached to the top portion **210** and the cannula **214** is attached to the bottom portion **212**. Inside the bottom portion **212** is a septum **222** in which the introduction needle **216** pierces through in the starting position, which is shown in, for example, FIGS. **4A-4E**. In various embodiments, the infusion device system **218** may also include a fluid source which in some embodiments may be an infusion pump apparatus which includes a disposable housing assembly **202** and a reusable housing assembly **200**. In various embodiments, the disposable housing assembly **202** includes a reservoir **518** and an infusion device mating assembly **204**.

(28) In practice, the infusion device **206** is pushed into the subcutaneous skin layer of a user by applying a downward force to the top portion **210** of the infusion device **206**. The introduction needle **216** pierces the skin and provides entry for the cannula **214**. Once this is completed, the top portion **210** may be removed from the bottom portion **212** by applying an opposite force onto the top portion **210**. The top portion **210** and introduction needle **216** are then removed from the bottom portion **212** and the user.

(29) In various embodiments, the infusion device **206** may be inserted into a user through a disposable housing assembly **202** infusion device mating assembly **204**. In these embodiments, the infusion device **206** is connected to the infusion device mating assembly **204** and the bottom portion **212** of the infusion device **206** is attached to the infusion device mating assembly **204**. The attachment mechanism may be any mechanism, and in some embodiments, may include one or more locking fingers **224**, **226** which, in various embodiments, may be spring loaded and when the bottom portion **212** is placed onto the infusion device mating assembly **204**, the locking fingers **224**, **226** lock into receiving areas **228**, **230** on the infusion device mating assembly **204** configured to receive and retain the locking fingers **224**, **226**. In various embodiments, the receiving areas **228**, **230** may be grooves, however, in various other embodiments, the receiving areas **228**, **230** may vary and in various embodiments including other embodiments of attachment mechanisms, the receiving features may vary. The top portion **210** may then be separated from the bottom portion **212** by application of force in the direction away from the infusion device mating assembly **204**. When the top portion **210** is separated from the bottom portion **212**, the introduction needle **216** is removed from the cannula **214**, leaving the cannula **214** inserted inside the user (or not inserted into a user).

(30) In various embodiments, attaching the bottom portion **212** to the infusion device mating

assembly **204** forms a fluid connection between the reservoir **518** in a disposable housing assembly **202** and the cannula **214**.

(31) In various embodiments of the infusion device system **218**, the disposable housing assembly **202** may include a viewing opening **208** located adjacent to the infusion device mating assembly **204** allowing the user or a caregiver to view the cannula **214**. This may be desirable/beneficial for many reasons, including, but not limited to, the ability to view the status of the cannula **214** that is inserted in the user. This may be desirable/beneficial for many reasons, including, but not limited to, the ability to determine whether the cannula **214** has become dislodged or if there is blood or other indication of a potential occlusion within the cannula.

(32) Although the embodiments shown in FIGS. **4A-4K** show an infusion device **206** inserted at an angle, and the infusion device mating assembly **204** is also at an angle with respect to the disposable housing assembly **202** and reusable housing assembly **200**, in various embodiments, the infusion device mating assembly **204**, disposable housing assembly **202** and reusable housing assembly **200** may be configured such that the infusion device **206** is inserted perpendicular to the infusion device mating assembly **204**, disposable housing assembly **202** and reusable housing assembly **200**.

(33) Referring now also to FIGS. **5A-5F** in various embodiments, the infusion device system **318** may include an infusion device **306** which in various embodiments may include a top portion **310**, a bottom portion **312**, a cannula **314** and an introduction needle **316**. The introduction needle **316** is attached to the top portion **310** and the cannula **314** is attached to the bottom portion **312**. Inside the bottom portion **312** is a septum **322** in which the introduction needle **316** pierces through in the starting position, which is shown in, for example, FIGS. **5A-5B**. In various embodiments, the infusion device system **318** may also include a fluid source which in some embodiments may be an infusion pump apparatus which includes a disposable housing assembly **302** and a reusable housing assembly **300**. In various embodiments, the disposable housing assembly **302** includes a reservoir **518** and an infusion device mating assembly **304**.

(34) In practice, the infusion device **306** is pushed into the subcutaneous skin layer of a user by applying a downward force to the top portion **310** of the infusion device **306**. The introduction needle **316** pierces the skin and provides entry for the cannula **314**. Once this is completed, the top portion **310** may be removed from the bottom portion **312** by applying an opposite force onto the top portion **310**. The top portion **310** and introduction needle **316** are then removed from the bottom portion **312** and the user.

(35) In various embodiments, the infusion device **306** may be inserted into a user through a disposable housing assembly **302** infusion device mating assembly **304**. In these embodiments, the infusion device **306** is connected to the infusion device mating assembly **304** and the bottom portion **312** of the infusion device **306** is attached to the infusion device mating assembly **304**. The attachment mechanism may be any mechanism, and in some embodiments, may include one or more locking fingers **324**, **326** which, in various embodiments, may be spring loaded and when the bottom portion **312** is placed onto the infusion device mating assembly **304**, the locking fingers **324**, **326** lock into receiving areas **328**, **330** on the infusion device mating assembly **304** configured to receive and retain the locking fingers **324**, **326**. In various embodiments, the receiving areas **328**, **330** may be grooves, however, in various other embodiments, the receiving areas **328**, **330** may vary and in various embodiments including other embodiments of attachment mechanisms, the receiving features may vary. The top portion **310** may then be separated from the bottom portion **312** by application of force in the direction away from the infusion device mating assembly **304**. When the top portion **310** is separated from the bottom portion **312**, the introduction needle **316** is removed from the cannula **314**, leaving the cannula **314** inserted inside the user (or not inserted into a user).

(36) In various embodiments, attaching the bottom portion **312** to the infusion device mating assembly **304** forms a fluid connection between a reservoir **518** in a disposable housing assembly

302 and the cannula **314**.

(37) In various embodiments of the infusion device system **318**, the disposable housing assembly **302** may include a viewing opening (see for example, FIG. 4, item **208**) allowing the user or a caregiver to view the cannula **314**. This may be desirable/beneficial for many reasons, including, but not limited to, the ability to view the status of the cannula **314** that is inserted in the user. This may be desirable/beneficial for many reasons, including, but not limited to, the ability to determine whether the cannula **314** has become dislodged or if there is blood or other indication of a potential occlusion within the cannula.

(38) Although the embodiments shown in FIGS. 5A-5F show an infusion device **206** inserted perpendicular with respect to the disposable housing assembly **302**, the reusable housing assembly **300**, and the infusion device mating assembly **304**, in various embodiments, the infusion device mating assembly **304**, disposable housing assembly **302** and reusable housing assembly **300** may be configured such that the infusion device **306** is inserted at an angle to the infusion device mating assembly **304**, disposable housing assembly **302** and reusable housing assembly **300**.

(39) Referring now also to FIGS. 6A-6G in various embodiments the infusion device system may include an infusion device **406** which in various embodiments may include a top portion **412**, a bottom portion **426**, a cannula **414**, an introduction needle **416** and an adhesive strip **418**. The introduction needle **416** is attached to the top portion **412** and the cannula **414** is attached to the bottom portion **426**. In various embodiments, the top portion **412** may be configured such that it may be attached to an autoinsertor **420**. Although the size and shape of the top portion **412** may vary in various embodiments, in some embodiments, the size and shape is configured such that it may be used within an autoinsertor **420**. In various embodiments, inside the bottom portion **426** is a septum **422** in which the introduction needle **416** pierces through. In various embodiments, the infusion device system may also include a fluid source which in some embodiments may be an infusion pump apparatus which includes a disposable housing assembly **402** and a reusable housing assembly **400**. In various embodiments, the disposable housing assembly **402** includes a reservoir **518** and an infusion device mating assembly **404**.

(40) In practice, the infusion device **406** is pushed into the subcutaneous skin layer of a user by applying a downward force to the top portion **412** of the infusion device **406**. The introduction needle **416** pierces the skin and provides entry for the cannula **414**. Once this is completed, the introduction needle **416** may be removed from the bottom portion **412** by applying an opposite force onto the introduction needle **416**. The introduction needle **416** is then removed from the bottom portion **412** and the user. The adhesive strip **418** maintains the location of the bottom portion **418** on the skin of the user.

(41) In various embodiments, the downward force is generated by the autoinsertor **420** which, in various embodiments, includes a spring **424** and a mechanism that releases the spring **424** which forces a push plate **410** downward. This action drives the push plate **410** into the bottom portion **412**. Once the push plate **410** is connected with the bottom portion **412**, the autoinsertor **420** may be removed, leaving the bottom portion **412** of the inserter device **406** attached to the user by way of the adhesive strip **418**. In various embodiments, a mechanism may be used that pulls the push plate **410** back into the autoinsertor **420** and therefore, in these embodiments, the autoinsertor **420** serves as a container for the introduction needle **416**. In various embodiments, the adhesive strip **418** may be any size and shape, including, but not limited to, the size and shape shown in FIGS. 6A-6G.

(42) The disposable housing assembly **402** infusion device mating assembly **404** may be mated with the bottom portion **412** of the infusion device **406** by placing the infusion device mating assembly **404** on top of the bottom portion **412** and then applying force in the direction towards the bottom portion **412**. As can be seen for example in FIG. 6E, in various embodiments, the infusion device mating assembly **404** is shaped such that it may be placed over the bottom portion **412** in any orientation. This may be beneficial/desirable for many reasons, including, but not limited to,

the ability of a user/caregiver to vary the orientation of the disposable housing assembly **402** with respect to the bottom portion **412** of the infusion device. In various embodiments, the infusion device mating assembly **404** may be attached to the bottom portion **412** in any orientation, thus, the user may place the disposable housing assembly **402** in any orientation within 360 degrees relative to the bottom portion **412**.

(43) In various embodiments, the infusion device mating assembly **404** may include a feature configured to interact with a feature on the bottom portion **412** that secures the infusion device mating assembly **404** to the bottom portion **412**. In various embodiments, these features may include locking features including but not limited to, tongue and groove features. In various embodiments, attaching the bottom portion **412** to the infusion device mating assembly **404** forms a fluid connection between a reservoir **518** in a disposable housing assembly **402** and the cannula **414**.

(44) In various embodiments of this embodiment of the infusion device system, if and when a user wishes to move the location of the disposable housing assembly **402**, the infusion device mating assembly **404** may be removed from the bottom portion **412** and the bottom portion **412** and cannula **414** may be removed from the user. Following, the user may connect another infusion device **406** bottom portion **412** to their skin in a different location on their body and reconnect the disposable housing assembly **402**. In some embodiments, and referring also to FIG. **10**, rather than removing the bottom portion **412** and the cannula **414**, the infusion device system may include a predetermined length of tubing **428** or other that connects to the bottom portion **412** on a first end of the tubing and to the infusion device mating assembly **404** on the second end of the tubing, thereby creating a fluid connection between the disposable housing assembly **402** and the cannula **414** without the need for removing a first cannula and inserting a second cannula.

(45) In various embodiments of the bottom portion, the bottom portion may be made from clear plastic or another clear material allowing for the cannula site to be viewed. This may be desirable/beneficial for many reasons, including, but not limited to, the ability to determine whether the cannula has become dislodged or if there is blood or other indication of a potential occlusion within the cannula. In various embodiments, tubing may be used in conjunction with any embodiment of the infusion device described herein. Also, the tubing, in various embodiments, may be any size and length.

(46) In various embodiments of the infusion device system, the disposable housing assembly **402** may include a viewing opening (see for example, FIG. **4**, item **208**) allowing the user or a caregiver to view the cannula **414**. This may be desirable/beneficial for many reasons, including, but not limited to, the ability to view the status of the cannula **414** that is inserted in the user. This may be desirable/beneficial for many reasons, including, but not limited to, the ability to determine whether the cannula **414** has become dislodged or if there is blood or other indication of a potential occlusion within the cannula.

(47) In various embodiments, the disposable housing assembly may include a fluid connector assembly, for example, one or more of the embodiments described U.S. patent application Ser. No. 13/788,260, filed Mar. 7, 2013 and entitled Infusion Pump Assembly, now U.S. Publication No. US-2014-0107579, published Apr. 17, 2014 which is hereby incorporated herein by reference in its entirety. Thus, in these embodiments, the infusion device mating assembly may be part of the fluid connector assembly. In various embodiments including a fluid connector assembly and a viewing opening, the viewing opening may be included on the fluid connector assembly.

(48) In any of the embodiments of the infusion device described herein the infusion device may connect to a length of tubing which is connected to a fluid source. However, in any of the embodiments of the infusion device described herein, the tubing may be optional and therefore, the infusion device is a tubless infusion device until and unless a user attaches a predetermined length of tubing to the infusion device.

(49) In various embodiments, these methods may be used with respect to any device and/or medical

device and/or any controller and/or remote controller for any device and/or medical device and/or any device used in conjunction with or in association with any device and/or medical device.

(50) A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made. Accordingly, other embodiments are within the scope of the following claims.

(51) While the principles of the invention have been described herein, it is to be understood by those skilled in the art that this description is made only by way of example and not as a limitation as to the scope of the invention. Other embodiments are contemplated within the scope of the present invention in addition to the exemplary embodiments shown and described herein.

Modifications and substitutions by one of ordinary skill in the art are considered to be within the scope of the present invention.

Claims

1. An infusion device system comprising: a housing assembly, the housing assembly comprising: a reservoir containing infusible fluid; a pump assembly; and a mating assembly; an infusion device, the infusion device comprising: a top portion having an introduction needle; and a bottom portion configured to attach to the mating assembly of the housing assembly, the bottom portion having a cannula; wherein the introduction needle is configured to pierce skin of a user and provide entry of the cannula; wherein the housing assembly defines a viewing opening that allows viewing, through the viewing opening, the portion of the cannula that has been inserted into the user; and wherein the cannula becomes fluidly connected to the reservoir after the bottom portion of the infusion device is attached to the mating assembly.
2. The infusion device system of claim 1, wherein the bottom portion of the infusion device further having a septum, and wherein the introduction needle is configured to pierce the septum when providing entry of the cannula into the skin of the user.
3. The infusion device system of claim 1, wherein the top portion of the infusion device is removably attached to the bottom portion of the infusion device and configured to be removed from the bottom portion after the cannula enters a subcutaneous skin layer of the patient.
4. The infusion device system of claim 1, wherein the mating assembly is configured to receive the infusion device at an angle.
5. The infusion device system of claim 1, wherein the viewing opening is arranged between the reservoir and the mating assembly.
6. The infusion device system of claim 1, wherein the housing assembly further comprises a volume sensor assembly.
7. The infusion device system of claim 1, wherein the bottom portion of the infusion device includes one or more locking fingers configured to lock into one or more receiving areas on the mating assembly.
8. The infusion device system of claim 7, wherein the top portion of the infusion device is removably attached to the bottom portion of the infusion device and configured to be removed from the bottom portion after the one or more locking fingers lock into the one or more receiving areas on the mating assembly.
9. The infusion device system of claim 1, wherein the mating assembly extends at an angle with respect to a portion of the housing assembly defining the viewing opening.
10. A housing assembly of an infusion device system, the housing assembly comprising: a reservoir containing infusible fluid; a pump assembly; and a mating assembly configured to receive an infusion device; wherein the housing assembly defines a viewing opening that allows viewing, through the viewing opening, of the portion of a cannula that has been inserted into a user with the infusion device through the mating assembly; and wherein the mating assembly is configured such that the cannula becomes fluidly connected to the reservoir after the infusion device is attached to

the mating assembly.

11. The housing assembly of claim 10, wherein the mating assembly is configured to receive the infusion device at an angle.

12. The housing assembly of claim 10, wherein the viewing opening is arranged between the reservoir and the mating assembly.

13. The housing assembly of claim 10, wherein the housing assembly further comprises a volume sensor assembly.

14. The housing assembly of claim 10, wherein the mating assembly defines one or more receiving areas configured to receive one or more locking fingers of the infusion device.

15. The housing assembly of claim 10, wherein the mating assembly extends at an angle with respect to a portion of the housing assembly defining the viewing opening.
